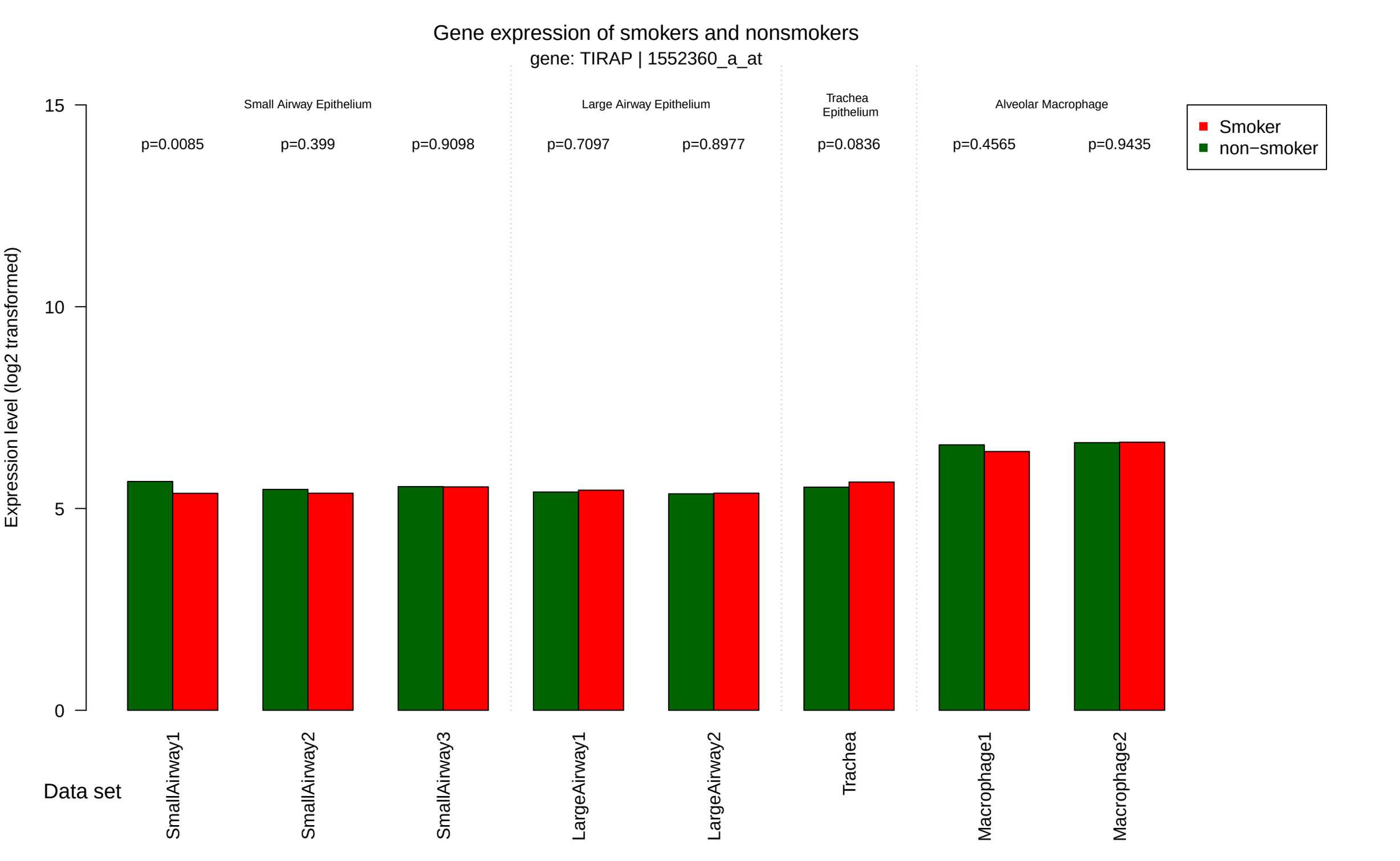
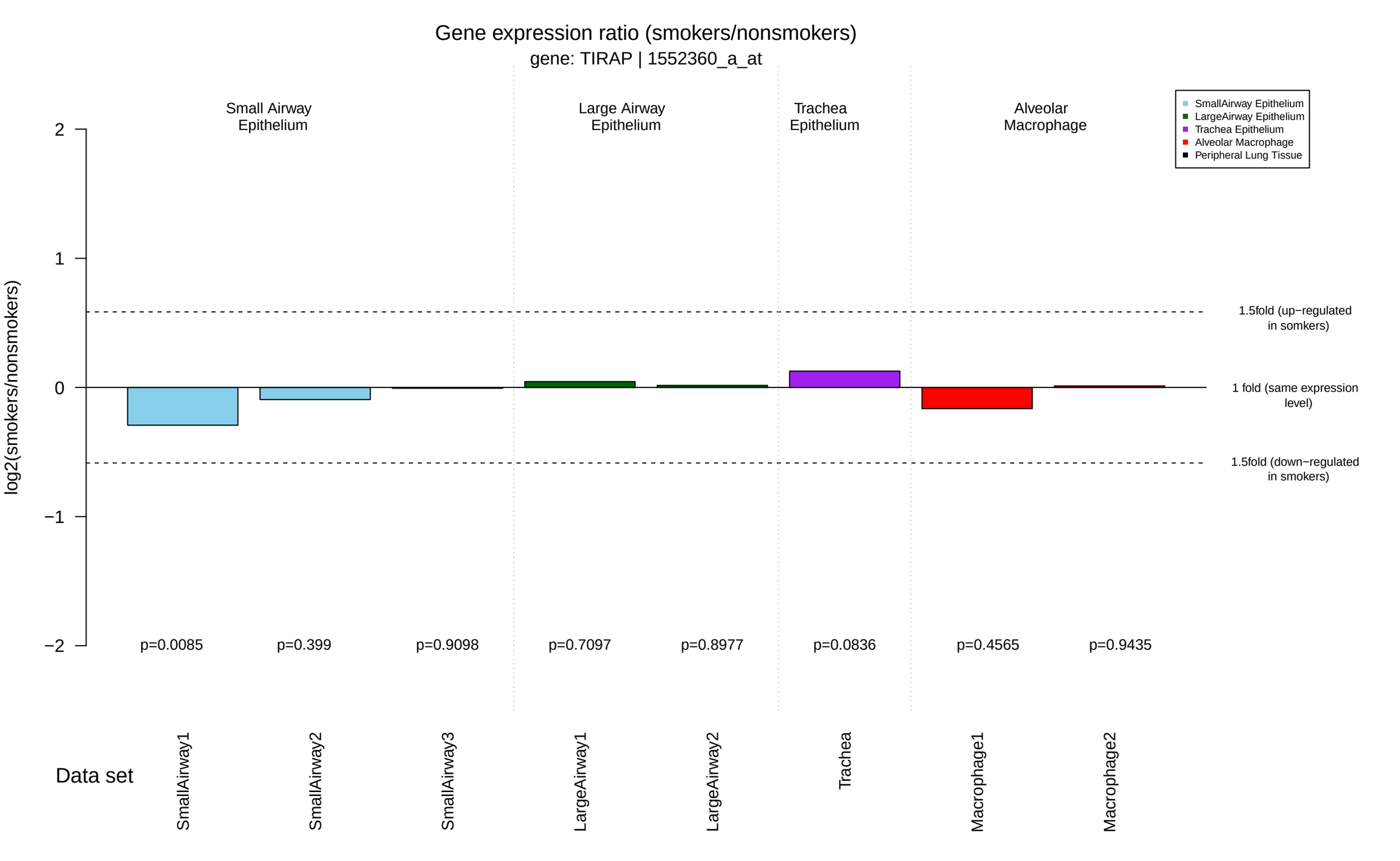
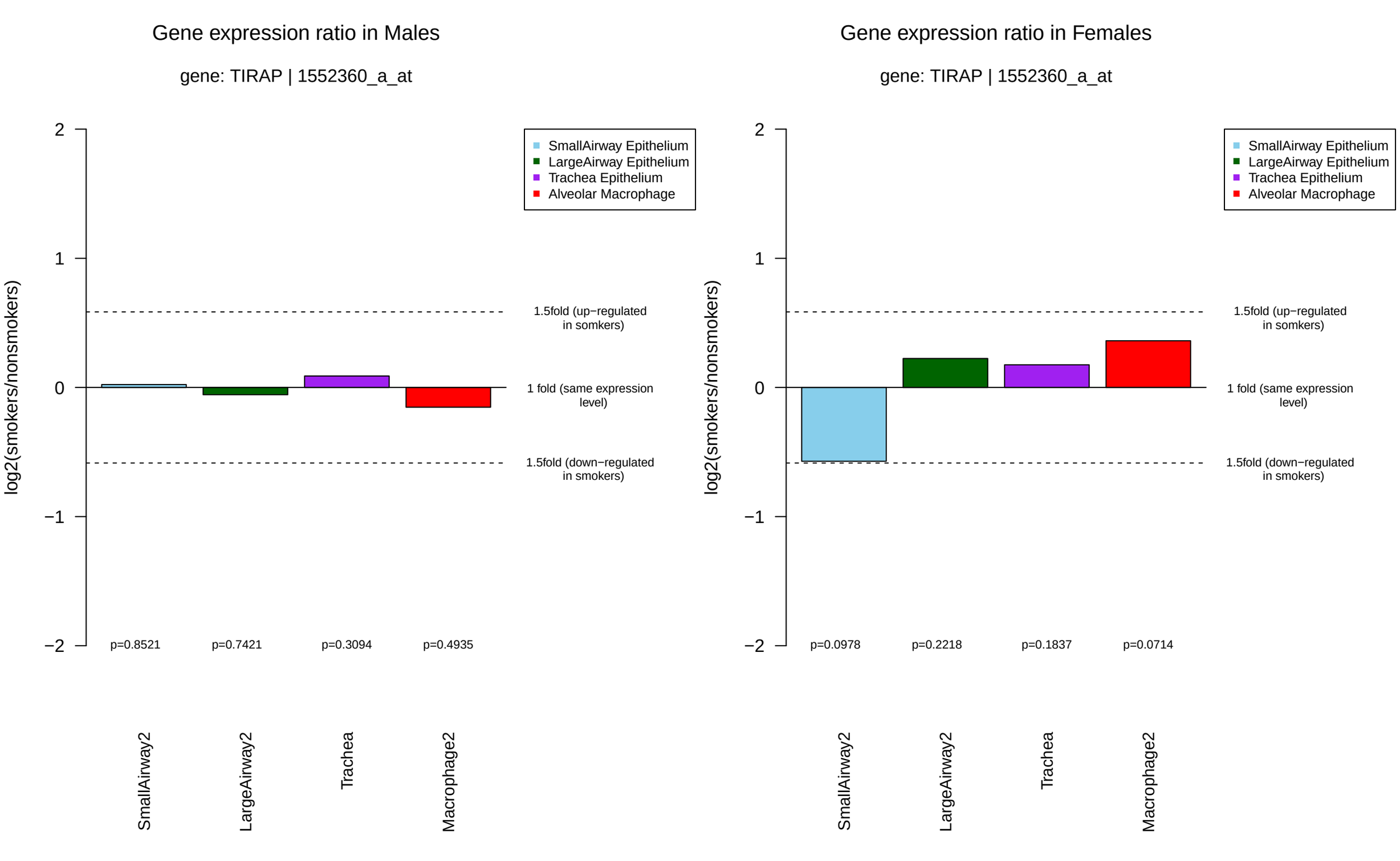
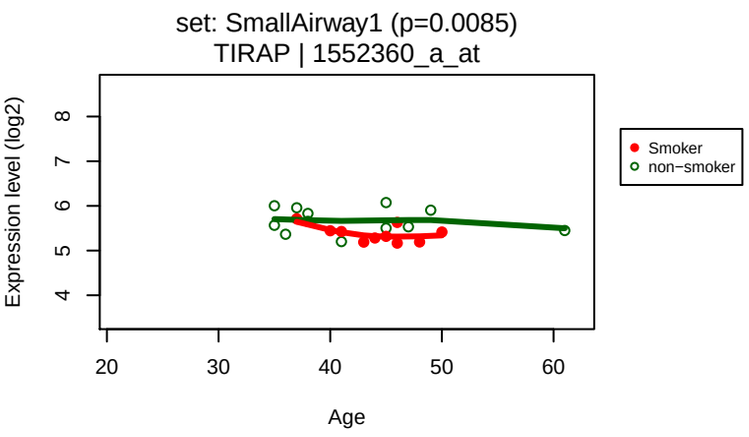
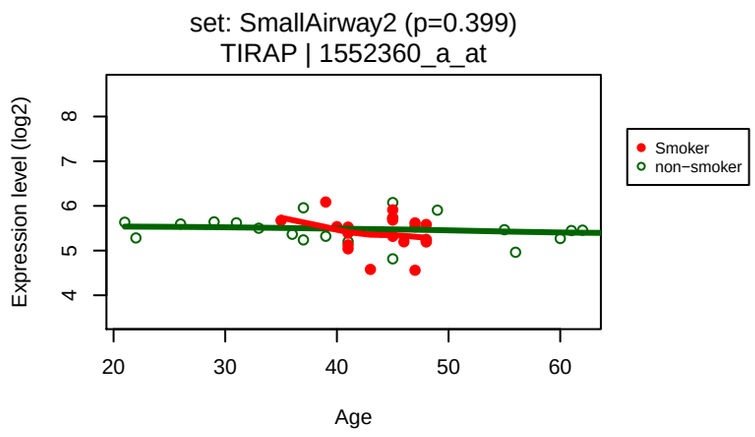
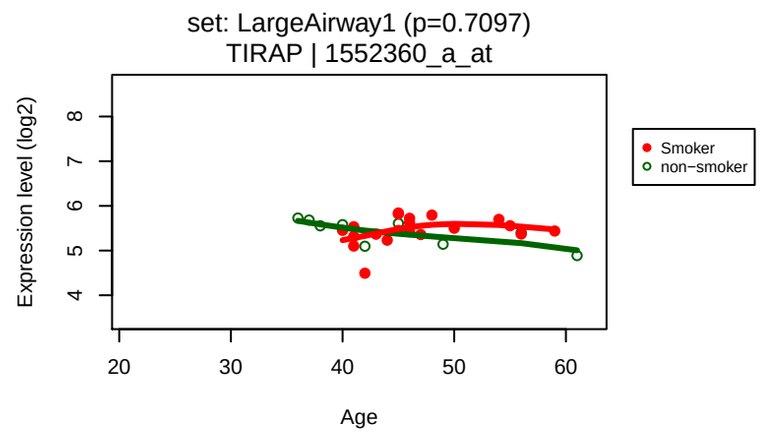

Gene expression of smokers and nonsmokers
gene: TIRAP | 1552360_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0085
p=0.399
p=0.9098
p=0.7097
p=0.8977
p=0.0836
p=0.4565
p=0.9435
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

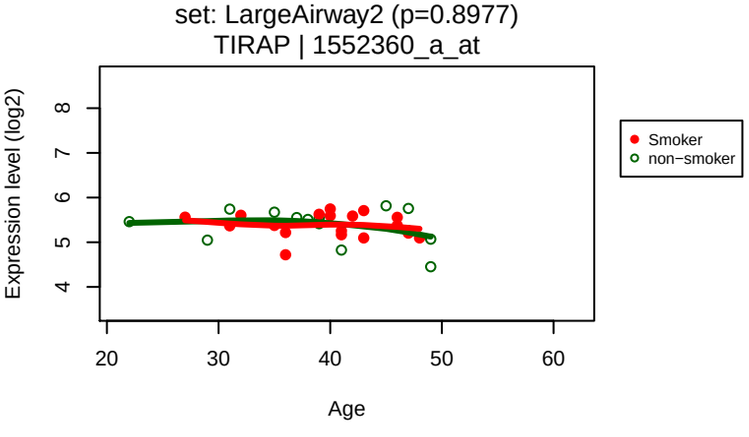

Gene expression ratio (smokers/nonsmokers)
gene: TIRAP | 1552360_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0085
p=0.399
p=0.9098
p=0.7097
p=0.8977
p=0.0836
p=0.4565
p=0.9435
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

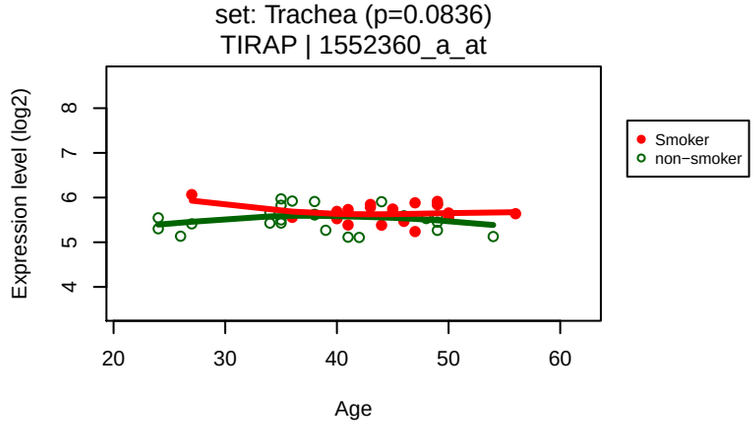

Gene expression ratio in Males
gene: TIRAP | 1552360_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8521
p=0.7421
p=0.3094
p=0.4935
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TIRAP | 1552360_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0978
p=0.2218
p=0.1837
p=0.0714
SmallAirway2
LargeAirway2
Trachea
Macrophage2

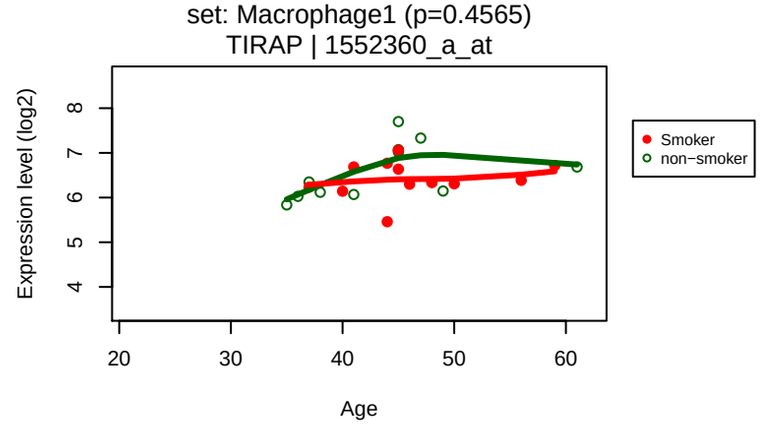

set: SmallAirway1 (p=0.0085)
TIRAP | 1552360_a_at
Expression level (log2)
Age
Smoker
non−smoker

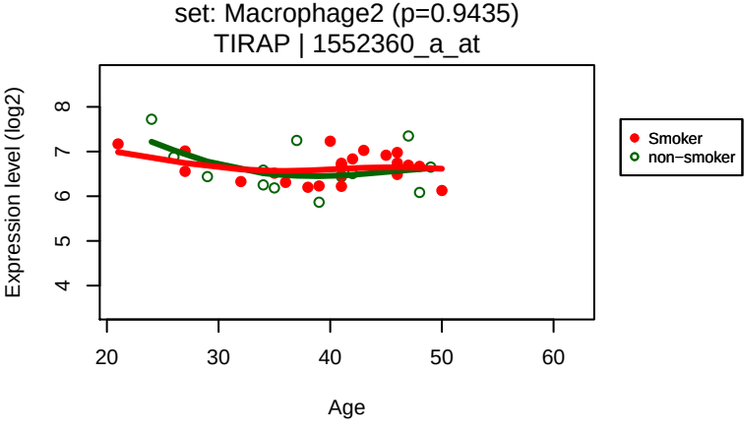

set: SmallAirway2 (p=0.399)
TIRAP | 1552360_a_at
Expression level (log2)
Age
Smoker
non−smoker

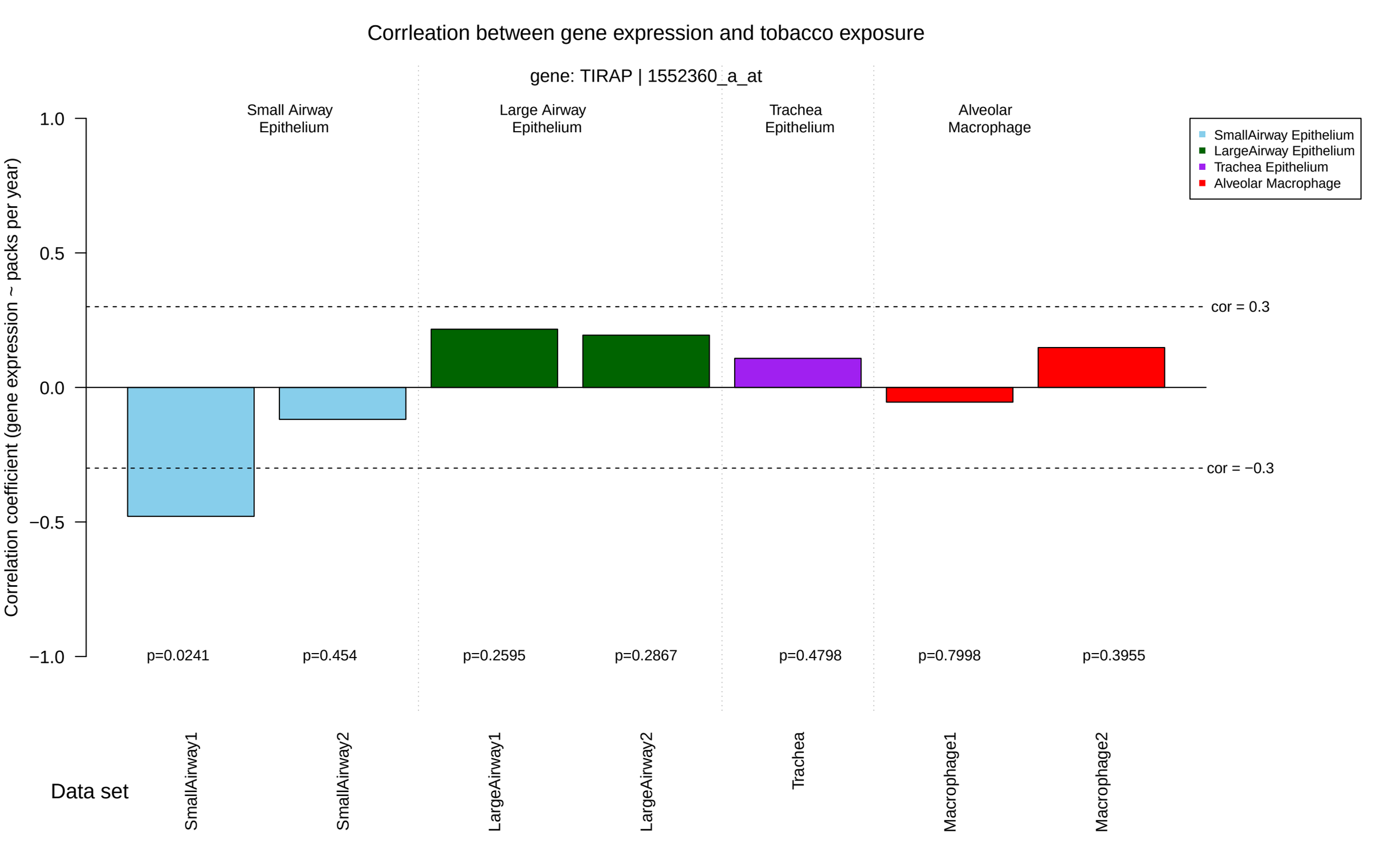

set: LargeAirway1 (p=0.7097)
TIRAP | 1552360_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8977)
TIRAP | 1552360_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0836)
TIRAP | 1552360_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4565)
TIRAP | 1552360_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9435)
TIRAP | 1552360_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TIRAP | 1552360_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0241
p=0.454
p=0.2595
p=0.2867
p=0.4798
p=0.7998
p=0.3955
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2